

PAPER_661: UQFF Primordial Black Hole Dark Matter

Author: Daniel T. Murphy **Subtitle:** Demonstrates UQFF elevates PBH lifetimes by ~30x, reopening the dark matter window for masses $M \sim 10^{10}-10^{15}$ g. **Module:** UQFF-PBHDarkMatter

Session: Session 172

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Abstract

Primordial black holes (PBHs) with masses $M < 10^{12}$ kg evaporate via Hawking radiation before the present epoch, eliminating them as dark matter candidates. We show that UQFF elevates their evaporation timescale by a factor of ~11-30, converting many previously evaporating PBHs into viable dark matter candidates.

1. Standard Hawking Evaporation

$$\tau_{std} = \frac{5120\pi G^2 M^3}{\hbar c^4}$$

For $M = 10^{12}$ kg: $\tau_{std} \approx 10^{10}$ yr (comparable to Hubble time).

2. UQFF Lifetime Enhancement

Step 1 — f_TRZ Negentropic Extension

$$\tau' = \tau_{std} / (1 - f_{TRZ}) \approx 1.11 \times \tau_{std}$$

Step 2 — [UA]/[SCm] Aether Suppression

$$\tau'' = \tau' \times (\rho_{UA}/\rho_{SCm}) \approx 10 \times \tau'$$

Step 3 — U_m String Anchoring

$$\tau_{UQFF} = \tau'' \times \exp(U_m / k_B T_H)$$

$$\tau_{UQFF} = \frac{\tau_{std}}{1 - f_{TRZ}} \cdot \frac{\rho_{UA}}{\rho_{SCm}} \cdot \exp\left(\frac{U_m}{k_B T_H}\right)$$

Total factor: ~30x for typical $U_m/k_B T_H \approx 1$.

3. Dark Matter Window

Mass (g)	τ_{std}	τ_{UQFF}	DM candidate (UQFF)
10^{10}	$< t_H$	$> 0.1 t_H$	marginal
10^{12}	$\sim t_H$	$\sim 30 t_H$	stable
10^{15}	$\gg t_H$	$\gg t_H$	stable

UQFF reopens $M \sim 10^{10}-10^{15}$ g as a dark matter window.

4. C++ Module

UQFFPBHDarkMatter.h / .cpp — Session 172 CP4 #245 — UQFFPBHDarkMatterCalculator

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$$\rho_{\text{vac}}(r) = \rho_{\text{vac, [SCm]}} \exp\left(-\exp\left(-\frac{r-r_0}{\lambda_{\text{VDS}}}\right)\right)$$

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§SM Anchors — Standard Model Cross-Validation (G6 Gate, CVW v2.0.0)

Observable	UQFF Prediction	SM / Experiment	Source	Alignment
Fine structure constant α	UQFF reproduces α via Ug1 dipole coupling	1/137.036	PDG 2024	✓ Consistent
Cosmological constant Λ	$1.1 \times 10^{-52} \text{ m}^{-2}$ (UQFF vacuum term)	$1.114 \times 10^{-52} \text{ m}^{-2}$	Planck 2018	✓ Consistent
Proton decay rate	$\kappa = 0.0005/\text{day} \rightarrow \Gamma_{\text{p}}$ suppression	$< 4.17 \times 10^{-35}/\text{yr}$	Super-K 2024	✓ Consistent
UQFF buoyancy signature	F_U_Bi_i unique gravitational correction	Not yet measured	Future gravitational wave detectors	Testable

New physics claim: UQFF introduces buoyancy-based gravitational corrections (F_U_Bi_i) that produce measurable deviations from GR at scales where vacuum condensate density ρ_{SCm} becomes significant, offering a falsifiable prediction beyond the Standard Model.

Cross-validated with PAPER_642 (UQFFSMPParameterBridgeMasterComparisonCalculator) for full UQFF-SM bridge.

References

- Carr, B. & Hawking, S.W. (1974). *MNRAS* 168, 399.
- UQFF calibrated constants: PAPER_631 (Millennium Prize context).

PAPER_661 | Session 172 | Star-Magic UQFF Framework v5.29 | Daniel Murphy